

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY, CHANGA  
M Sc (BIOCHEMISTRY)  
SEMESTER II EXAMINATION  
BC706: ENZYMOLOGY

Date: 29/04/2010

Time: 10.00 am to 1.00 pm

Total Marks: 70 minutes

**Instructions to the candidates:**

1. Attempt all questions
2. Questions in **Part A** should be answered in the question paper itself. Please do not write your candidate ID number on the question paper.
3. Write answers for Section I and Section II of **Part B** in separate answer sheets.

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**PART A**

Duration: 30 minutes

Total marks: 20

1. The enzymes that essentially catalyze the same reaction but differ in their structure are known as \_\_\_\_\_.
2. Kinetics of immobilized enzyme differs significantly from that of the same enzyme in free solution. True / False
3. For efficient enzyme catalysis, complementarity between substrate and active site is more preferred than complementarity between transition state and active site. True/False
4. The molecules that are tightly bound to the enzymes and are essential for its catalysis are known as \_\_\_\_\_.
5. Which of the following factors is **NOT** desirable while choosing the carrier matrices for immobilization?  
(a) Low cost (b) Inert nature  
(c) Highly reactive (d) Easy to discard
6. An enzyme with an EC number of 6.2.12.1 would belong to which broad class of enzymes?  
(a) Lyase (b) Ligase  
(c) Hydrolase (d) Isomerase
7. Which of the following is the mode of regulation employed by trypsin?  
(a) Allosteric activation (b) Occurrence as multienzyme complex  
(c) Occurrence as zymogen (d) Presence of isozymes
8. Which of the following kinetic constant best describe the rate of catalysis?  
(a)  $V_{max}$  (b)  $K_M$   
(c)  $K_{cat}$  (d)  $E_T$
9. Which of the following reactions does **NOT** involve the formation of ternary complex intermediate?  
(a) Ping-pong reactions (b) Single displacement reactions  
(c) Ordered sequential reactions (d) Random sequential reactions



10. Which of the following is **NOT** an assumption in deriving Michaelis-Menten's equation?
- (a)  $ES \rightarrow E + P$  is reversible (b)  $k_{-1} \gg k_2$   
 (c)  $[S] \gg [E]$  (d)  $[E_T] = [E] + [ES]$
11. Which of the following events marks the transition state stabilization by chymotrypsin?
- (a) Irreversible inactivation by DIPF (b) stabilization of oxyanion  
 (c) Formation of catalytic triad (d) None of the above
12. Which one of the following is **TRUE** for enzyme catalyzed reactions?
- (a) Free energy decreases, entropy decreases (b) Free energy increases, entropy decreases  
 (c) Free energy decreases, entropy increases (d) Free energy increases, entropy increases
13. Which one of the following best describes the inhibition of invertase by aniline, when immobilized on hydrophilic surface?
- (a)  $K_i$  will decrease (b)  $K_i$  will increase  
 (c)  $K_i$  remains unaltered (d) No inhibition
14. Which of the following is **TRUE** for single cell enzymatic studies?
- (a) Follows Michaelis-Menten kinetics (b) Enzymes exhibit dynamic disorder  
 (c) Turnover number is monitored (d) All of the above
15. Which of the following is an allosteric effector for hemoglobin?
- (a) CO (b) NO  
 (c) O<sub>2</sub> (d) 2,3-BPG
16. Michaelis-Menten's equation generally applies to which of the following enzymatic reactions?
- (a) Uni-substrate reactions (b) Bi-substrate reactions  
 (c) Multi-substrate reactions (d) All of the above
17. Which of the following inhibitions can be overcome increasing the substrate concentration?
- (a) Substrate inhibition (b) Product inhibition  
 (c) Non-competitive inhibition (d) Competitive inhibition
18. Which of the following is **TRUE** for non-competitive inhibition?
- (a) It is a special case of mixed inhibition (b) Inhibitor binds at enzyme's active site  
 (c) Inhibitor binds with equal affinity to [E] and [ES] (d)  $V_{max}^{app}$  reduces;  $K_M^{app}$  unaltered
19. Which of the following is **NOT TRUE** for co-operativity in Hemoglobin?
- (a) It demonstrates positive co-operativity  
 (b) It shows sigmoidal binding  
 (c) At lower  $pO_2$  all Hb subunits simultaneously bind to O<sub>2</sub>  
 (d) T state has lower affinity for O<sub>2</sub> as compared to R state
20. Which one of the following is **NOT TRUE** for Phillips mechanism of lysozyme catalysis?
- (a) Cleaves  $\beta$  (1,4) glycosidic linkages from NAM to NAG  
 (b) Reaction proceeds via a covalent intermediate  
 (c) Glu35 at active site acts as an acid  
 (d) Glucose ring is distorted from half chair to chair conformation



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**Part B**

**Total Marks: 50**

**Section I**

- Q.1**
- (A) Diagrammatically depict the microenvironment and macroenvironment of an immobilized enzyme particle. 2
- (B) Explain how the use of multipoint interactions for enzyme stabilization may alter the kinetics of immobilized enzymes. 3
- OR**
- (B) Describe the use of fluorescence based assays of single enzyme molecules using suitable example. 3
- (C) **Write short notes on (ANY TWO)** 5
- (i) Advantages of immobilized enzymes
  - (ii) Immobilized enzymes in clinical diagnostics
  - (iii) Membrane confinement mechanism of immobilization
- Q.2**
- (A) **Explain the following in brief (ANY TWO)** 4
- (i) Multienzyme complex
  - (ii) Factors affecting rates of enzyme catalyzed reaction
  - (iii) Interconvertible enzyme cycles
  - (iv) Effect of 2,3 Bisphosphoglycerate binding on hemoglobin function
- (B) (i) An arbitrary enzyme A has a  $V_{max}$  of  $3.5 \mu\text{moles l}^{-1}\text{min}^{-1}$  and  $K_M$  of  $0.62 \mu\text{moles}$ . In the presence of an inhibitor I used at the concentration of  $10 \mu\text{M}$ , the  $V_{max}^{app}$  was  $1.5 \mu\text{moles l}^{-1}\text{min}^{-1}$  and  $K_M^{app}$  was  $0.2 \mu\text{moles}$ .  
• Comment on the type of inhibition 2
- (ii) Write a note on competitive inhibition. 2
- OR**
- (B) What is co-operativity? Describe the symmetry model of co-operativity for action of allosteric enzymes. Highlight the drawbacks of symmetry model. 4
- (C) Write the Hill equation for allosteric enzymes? Explain the significance of Hill Coefficient. 2



## Section II

- Q.3** (A) Describe the salient features of active site of an enzyme. Discuss in short **any two** methods to determine the active site structure. 3
- (B) Enlist the catalytic strategies used by the enzymes. Explain **any two** in detail. 3
- OR**
- (B) Explain the mechanism of concerted general acid-base catalysis using a suitable example. 3
- (C) **Explain the following in brief (ANY TWO)** 4
- (i) Role of binding energy in enzyme catalysis
  - (ii) Substrate specificities of chymotrypsin, trypsin and elastase
  - (iii) Mechanism of transition state stabilization by chymotrypsin.
  - (iv) Catalytic role of metal ion in carbonic anhydrase
- Q.4** (A) Give an account on the following statement, 'Enzyme presence in a reaction imposes greater catalytic effect than most catalysts'. 3
- (B) Comment on the stereospecificity and geometric specificity of enzyme action. 3
- OR**
- (B) Discuss the Koshland's 'Induced Fit' hypothesis of enzyme action. Explain how it is different from Fischer's 'Lock-and-Key' hypothesis. 3
- (C) **Write short notes on** 4
- (i) Enzyme nomenclature
  - (ii) Cofactors of enzymes
- OR**
- (C) Describe a suitable procedure for extraction and purification of a cytosolic enzyme A from given bacterial cell culture. Enzyme A is a 400 kDa homotetramer. It is a metal dependent enzyme and can specifically bind to biotin. 4
- Q.5** (A) Write the Michaelis-Menten's equation for uni-substrate enzyme kinetics. Explain the significance and importance of the kinetic constants  $K_M$ ,  $V_{max}$ ,  $K_{cat}$  and  $K_{cat}/K_M$ . 5
- OR**
- (A) Discuss about **any three** graphical ways of analyzing the kinetic data of the enzymatic reactions. 5
- (B) Explain how Lineweaver-Burk's plot can be used to distinguish between sequential kinetics and ping-pong kinetics. 3
- OR**
- (B) Outline the classification of bi-substrate reaction mechanisms. Brief the mechanism of double displacement reactions. 3
- (C) 'Steady state kinetics cannot unambiguously establish a reaction mechanism'. Comment on the statement. 2